D8. Evaluation Dataset and Methodology

Question set, labelling method, evaluation design and limitations.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Label distribution and scope

Element	Current value	Methodological consequence
Source reports	2 extracted JSON reports	Integration evidence only
Report-question pairs	16	Too small for stable generalisation estimates
Gold answers	12 Yes; 4 No; 0 Insufficient Evidence	Macro-F1 and abstention safety are limited
Evidence labels	Gold chunk IDs for structure-aware 420/60	Rank metrics are invalid after boundary changes
Decision threshold	70% positive answers	Reproducible POC score, not expert certification

Evaluation integrity

Questions were derived from the actual supplied reports and applied consistently to both reports. The harness stores per-report answer labels and evidence IDs, avoiding a synthetic task that bears no relation to the documents being assessed. [E4][E6]

The limitation is equally important: source-derived labels are not independent expert labels. The package therefore uses a release gate that correctly remains BLOCKED until a larger balanced, independently reviewed corpus exists. [E7][E10]

Dataset registry

Dataset	Reports	Questions	Role
Initial sample	sample_report_001.txt	12	Baseline/smoke benchmark [E1]
Real JSON proof set	2-report.json; 4-report.json	8 per report; 16 total	Source-derived integration evaluation [E4][E6]
Intermediate Markdown	2-report_intermediate.md; 4-report_intermediate.md	0	Extraction audit artifacts; not duplicate indexed reports [E6]

Question design

The real-report set asks whether reports contain detailed findings, high severity, open status, CWE classifications, remediation, an all-closed statement, timeline evidence, and a no-findings statement. The same eight question templates are applied to both reports. [E6]

Gold answers and gold chunk IDs are stored per report in the saved run configuration. This supports traceability but is not independent annotation: the question set and labels were derived from the same supplied reports. [E4]

Method

Layer	Method	Reported measures
Retrieval	Compare returned chunk IDs with gold chunk IDs	Precision@k, Recall@k, Hit Rate@k, MRR, nDCG@k [E5]
Question response	Compare structured answer with gold answer	Accuracy, per-class metrics, macro-F1, confusion matrix [E5]
Grounding	Check evidence IDs and lexical support in returned context	Grounding rate; CRAG measures [E5][E8]
Audit score	Count Yes answers and apply 70% threshold	Positive count, percentage, PASS/FAIL [E5]

Limitations and next dataset

The 16-pair real-report set has 12 Yes and 4 No gold labels and zero Insufficient Evidence examples. Its macro-F1 cannot establish abstention safety. [E5]

A credible next evaluation should use report-level train/dev/test separation, independent labels and acceptable evidence spans, multiple agencies and report types, and explicit missing-evidence, contradiction, OCR-noise and visual cases. The planning target is at least 40 reports across VAPT, network, mobile and web/application domains. [E8]

Conclusion

This dataset is suitable for implementation integration and error analysis. It is not sufficient for generalisation or production-quality claims.

Controlled-comparison contract

When comparing	Held constant	Changed variable
LLMs	Reports, 16 pairs, labels, retrieval budget, prompt schema, seed	Generator model
ICL	Reports, labels, retrieval, Qwen2.5 control, seed	Example-selection strategy
Prompts	Reports, labels, retrieval, Qwen2.5 control, seed	v1 direct versus v2 evidence review
Retrieval	Reports, labels, generator, top-8, prompt	Dense-only versus hybrid
Chunking	Reports, questions, answer labels and generator	Chunk boundaries and window parameters

Methodological judgement

This protocol is stronger than an illustrative chatbot demo because every comparison declares what was held constant and records a run-specific configuration and log. It permits a reviewer to distinguish a model change from a retrieval, prompt or data change. [E10]

The protocol still has a hard validity boundary: both the questions and gold labels are source-derived from two supplied reports. The evidence can support component selection for this POC; it cannot support claims about real-world accuracy, missing-evidence safety or generalisation across audit agencies. [E5][E10]

Label provenance and split policy

The current 16 records use the same eight benchmark-question templates across both supplied reports. This is useful for checking whether the integrated pipeline can apply a stable assessment rubric to more than one input, and the stored report/question IDs make every aggregate count reproducible. [E4][E6]

It is not a train/test split. The questions and labels were authored from the supplied reports, so there is a risk that the benchmark reflects obvious report wording rather than the more difficult expert judgement required in deployment. This is why the package labels the work an evaluated POC instead of a validated decision system. [E10]

The next dataset must be frozen before model comparison: separate reports by source at report level, have domain reviewers label the verdict and acceptable evidence spans, and reserve missing-evidence/contradiction cases. This prevents accidental leakage and permits credible per-class error reporting. [E8]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E5] results/production_real_reports/1787846335_de0ded86a7db/{retrieval_metrics,question_level_llm_metrics,sota_faithfulness_metrics,audit_level_scores}.json: scored metrics and audit decisions.
- [E6] sample_data/{2-report.json,4-report.json,real_reports_benchmark_questions.json}: source reports and eight source-derived questions per report.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.